Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Scam Shield

a. Team name: **Scam Shield**

b. Team leader name: **Naman Shrimal**

c. Problem Statement:
BFSI: Intelligent Risk, Fraud & Financial Experiences
People in India lose crores to scam messages, QR codes and calls that look real. Scam Shield lets anyone check what they were sent in seconds, then act: a complaint draft, a shared scam list that warns the next person, and a WhatsApp card for the family. Community benefit: fewer people lose their savings, and every report protects the next one.

Brief about the idea

Check what you were just sent before you act on it.

Free for anyone in India, and built for the scams people meet here: fake payment receipts, "digital arrest" calls and UPI QR tricks.

One multimodal Gemini call

Text, screenshots, QR codes, links and Hinglish voice notes are read together by Gemini on Vertex AI.

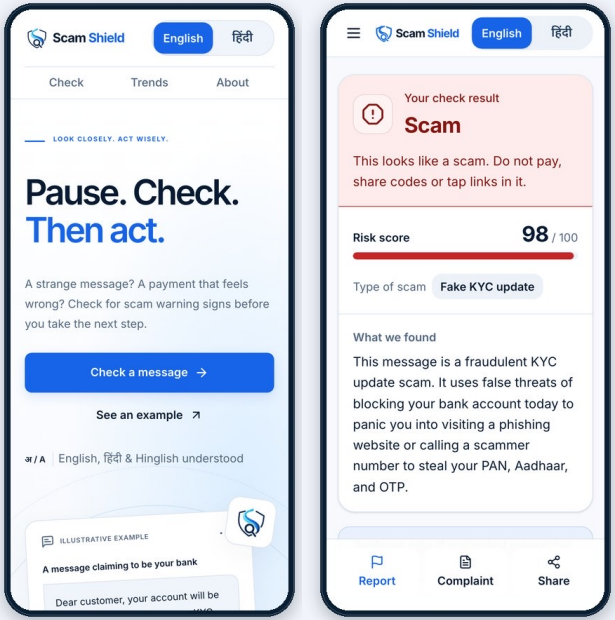
A verdict you can check

Scam, Suspicious or No scam signs found. Every red flag quotes the exact words. English or Hindi.

From verdict to action

A complaint draft for 1930 and cybercrime.gov.in, a shared scam list and a WhatsApp warning card.

Careful by design: links open in an isolated browser, never on your phone. It never says "safe": it gives guidance and you decide.



The live app, as seen on a phone.

Opportunities

Rs 19,813 crore reported lost to cyber fraud in 2025, across about 21.78 lakh complaints (I4C data, National Cyber Crime Reporting Portal)

How different is it from existing ideas?

Truecaller Scam Checker (Sep 2026): checks links, numbers and messages against community reports; screenshot upload is only planned.

Google scam detection: Messages on Pixel 6 and newer, and calls on Pixel 9 and newer, in India.

Chakshu (Sanchar Saathi): a reporting facility for suspicious calls, SMS and WhatsApp messages.

Scam Shield: reads the content itself, so it can flag scams nobody has reported yet, from text, screenshots, QR codes, links and voice notes.

How will it solve the problem?

- Checks before you pay, scan or click
- Quotes every red flag from the message
- Turns a scam into action: complaint, shared list, family warning

USP of the proposed solution

- Catches scams nobody has reported yet
- Made for India: fake payment receipts, "digital arrest" calls, UPI QR tricks
- Understands Hinglish voice notes

List of features offered by the solution

5 input types
Text, screenshot, QR code, link and voice note.

Clear verdict
Scam, Suspicious or No scam signs found, with a risk score.

Evidence quoted
Each red flag points to the exact words in the message.

English + Hindi
Hinglish understood; answers in the language you pick.

Complaint draft
Ready text for 1930 and cybercrime.gov.in.

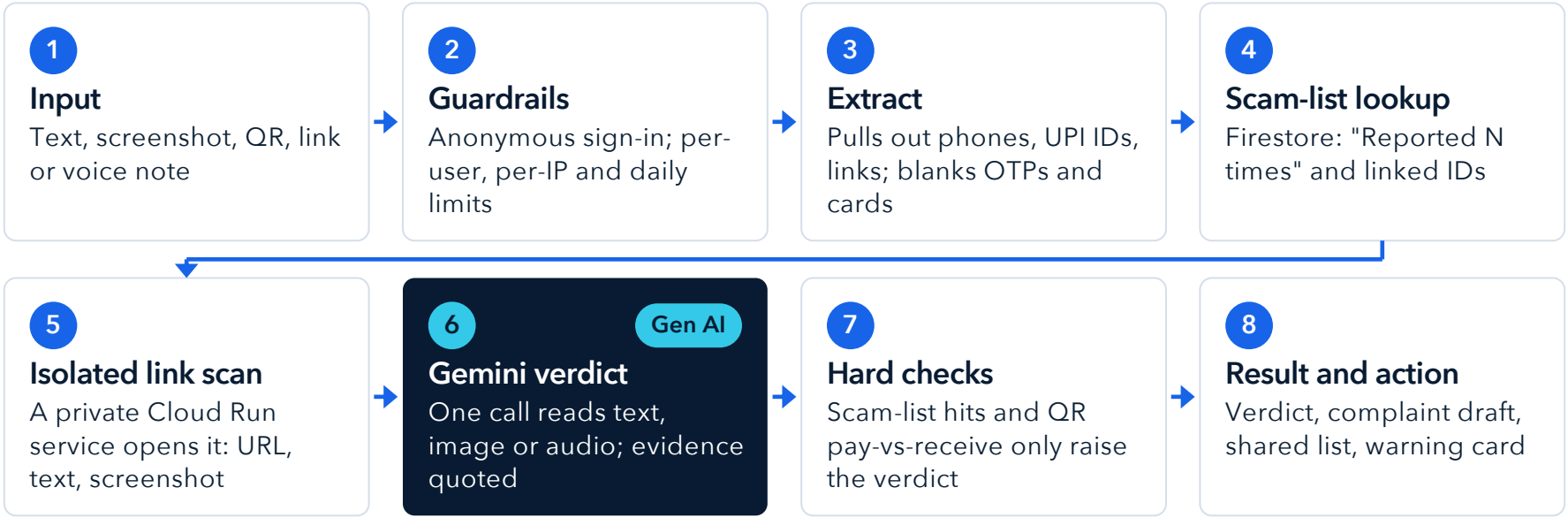
Shared scam list
"Reported N times", with linked numbers, UPI IDs and sites.

Warning card
One tap shares a warning with the family WhatsApp group.

Agent tools
WebMCP tools let AI agents check and report.

Also: trends, history, one-tap examples, an installable app and the Android share menu.

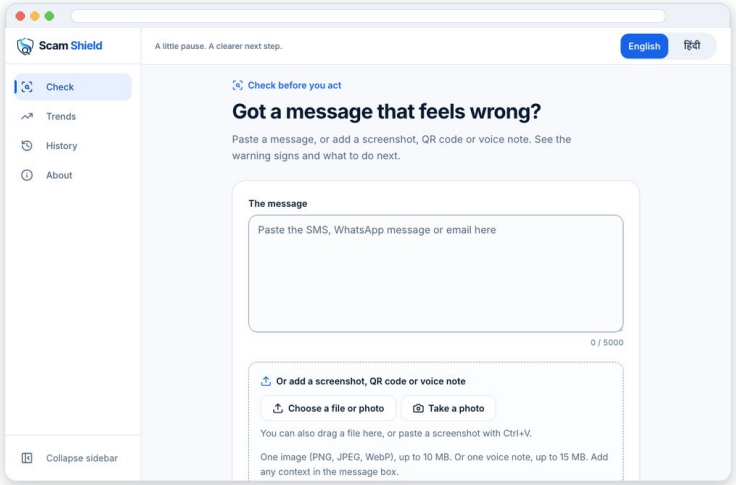
Process flow diagram or Use-case diagram



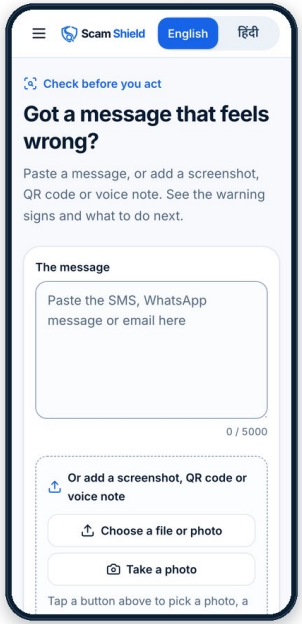
Why Gen AI: rules cannot read a screenshot or a Hinglish voice note, or judge a scam nobody has reported. Gemini does. The hard checks can only raise its verdict, never lower it.

If Gemini fails or times out, the person sees a clear retry message; no verdict is invented and nothing is stored.

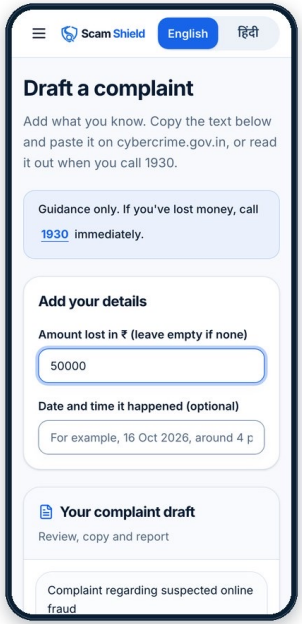
Wireframes/Mock diagrams of the proposed solution (optional)



Web app
The same flow on desktop, with a sidebar.



1 Paste or share
Text, photo, QR or voice note.

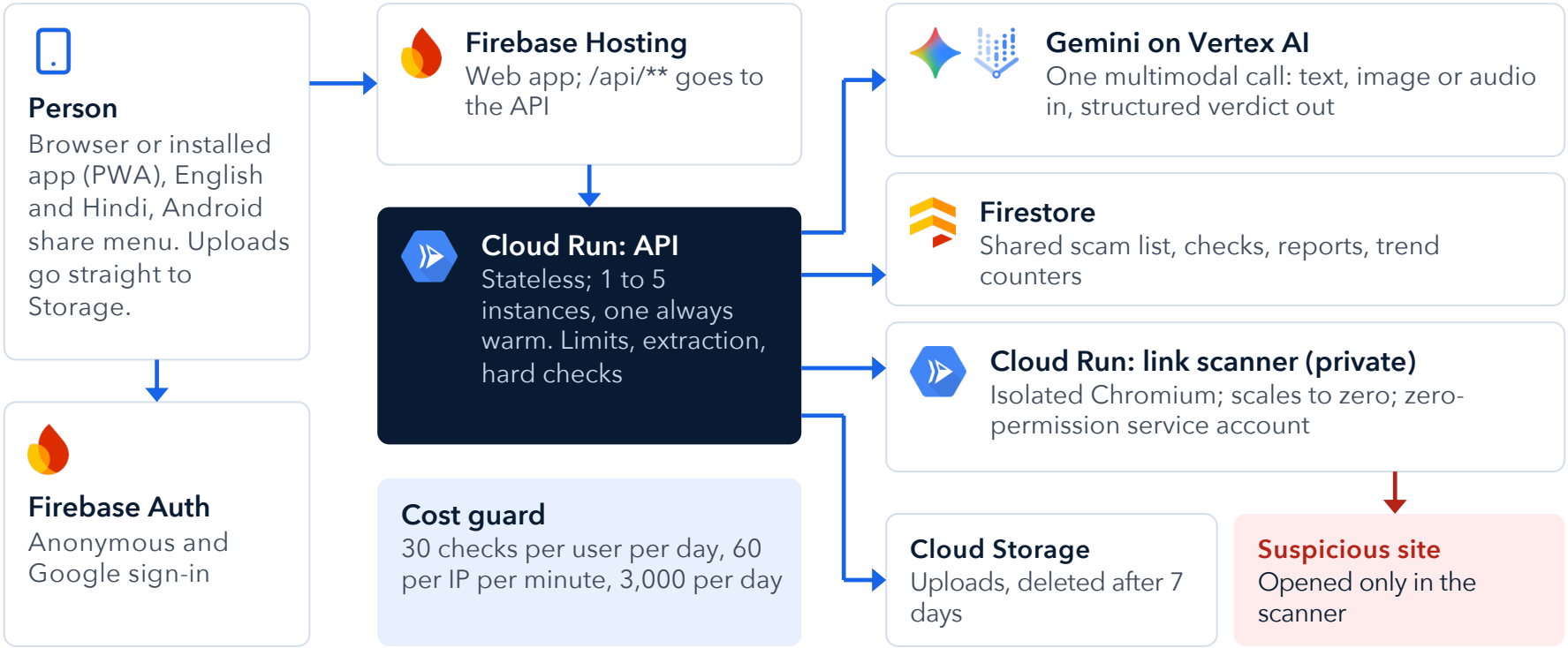


2 Act on a scam
A complaint draft ready to file.



3 Hindi
The whole app switches language.

Architecture diagram of the proposed solution



Product names and icons are trademarks of Google LLC, shown only to identify the technologies used. No endorsement or affiliation implied.

Technologies to be used in the solution

Gen AI that does real work

- **Gemini 3.7 Flash** on Vertex AI, with Gemini 2.5 Flash as automatic fallback
- **One multimodal call** reads text, screenshots and audio (Hinglish voice)
- **Schema-validated output;** every red flag quotes evidence; English or Hindi
- **Hard checks only raise** the verdict: scam list, QR pay-vs-receive
- **Does what rules cannot:** unreported scams, images, voice
- **WebMCP agent tools:** check_message and report_scam

Google Cloud and Firebase, built to scale

- **Cloud Run API:** stateless and autoscaling
- **Link scanner:** scales to zero, own zero-permission service account
- **Firestore** shared scam list; **Firebase** Hosting, Auth, Storage

Application

React 18 | Vite | Tailwind | Workbox PWA | Node 22 | TypeScript | Hono | zod | Playwright | sharp + jsQR

590+ automated tests, plus a 185-case live benchmark

Product names and icons are trademarks of Google LLC, shown only to identify the technologies used. No endorsement or affiliation implied.

Estimated implementation cost (optional)

FIXED

about US$13

per month (estimate)

One Cloud Run API instance kept warm (1 vCPU, 1 GiB, Mumbai) at US$0.0000025 per vCPU-second and per GiB-second, before any free tier.

PER CHECK

US$0.004–0.008

per text check (estimate)

Gemini 3.7 Flash at about 2,500 input and 500 to 1,500 output tokens. US$0.75 in, US$3.75 out per 1M tokens to 31 Dec 2026; double after.

FREE TIERS

Free tiers

expected at this volume (estimate)

Firebase Hosting, Auth, Firestore and Storage are expected to stay inside free tiers. The link scanner bills only while it scans.

Limits are the real control

30 checks per user per day, 60 per IP per minute, 3,000 per day overall. These cap spending.

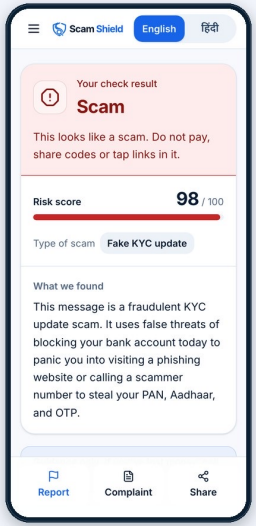
Ceiling at the daily cap

3,000 checks at the per-check estimate is about US$11 to 23 per day for Gemini.

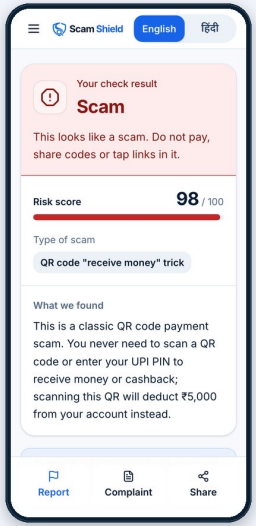
Budget alert: Rs 2,000 a month

Alerts at 50%, 90% and 100% (Rs 1,000, Rs 1,800, Rs 2,000). An alert notifies; it does not stop spending.

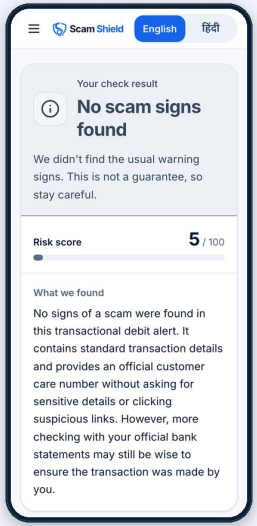
Snapshots of the prototype



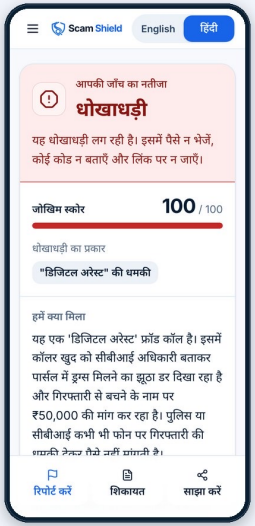
Scam
Fake KYC SMS, evidence quoted



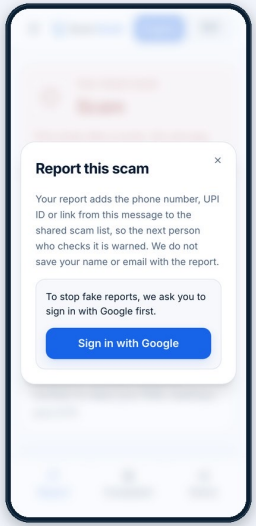
Scam
QR asks you to pay, text says receive



No scam signs found
A genuine bank alert; not a guarantee



Scam, in Hindi
Hinglish voice note, fake police call



Report
Adds it to the shared scam list

Live benchmark, 185 cases over 3 runs: 98.3% accuracy, 100% scam recall, 3.3% false positives (3 of 90 genuine)

Prototype Performance report/Benchmarking

98.3%	100%	3.3%	96.7%
Accuracy	Scam recall	False-positive rate	Scam type correct

185 synthetic cases over 3 live runs: 90 scams, 90 genuine, 5 ambiguous (not scored). Accuracy is 177 of 180 scored.

Run	Cases	Accuracy	Scam recall	False positives
v1 original	50	100%	100%	0%
v2 harder cases	35	93.3%	100%	12.5%
v3 realistic	100	99.0%	100%	2.0%
Combined	**185**	**98.3%**	**100%**	**3.3%**

Latency: text p50 4.5 s, p95 18.2 s; image p50 6.9 s; voice 10.3 s. **Errors:** 0. p50 is the median, p95 the slowest 5%.

Keyword baseline: on v2's 34 text cases it scored 94.1%, the same as Gemini, with different mistakes; v3 84.0% accuracy and 72.0% recall (Gemini 99.0% and 100%).

First run, 50 cases: 100% accuracy, 100% recall, 0% false positives. **Hindi run,** same 50 cases: 100%, 100%, 0%.

Read this honestly

- **185 cases, written by the authors:** a test, not a guarantee. 90 scams over 13 types, 90 genuine.
- **3 false positives:** scam-awareness forward that quotes a scam link, delivery update with a tracking link, friend asking for a short loan. All misses and false positives: eval/results/combined.md.
- **Why gemini-3.7-flash:** an earlier 50-case run on gemini-3.8-flash scored 98.0% accuracy with 4.2% false positives, image checks up to 34 s.

Additional Details/Future Development (if any)

More Indian languages
Tamil, Telugu, Marathi, Bengali and more.

Bank & telecom partners
Verified reports feed the shared scam list.

URL reputation feeds
Add Google Web Risk next to the isolated scan.

Android app
Share SMS and notifications to a check.

Google Search grounding
Switch on for link checks once quota allows.

Call-recording upload
Check recorded calls, not just voice notes.

Misinformation mode
The same engine for "is this forward real?"

Plans, not shipped.
Search grounding is built but off by default.

How it scales

Stateless API
Cloud Run adds instances (1 to 5 today) as traffic grows; the daily cap is a setting, not a rebuild.

Managed data
Firestore, Hosting, Auth and Storage need no servers to run; uploads auto-delete after 7 days.

Cost stays bounded
Per-user, per-IP and global limits rise only as budget allows; App Check adds abuse protection next.

Provide links to your:

- **GitHub Public Repository**
  https://github.com/Namans12/scam-shield
- **Demo Video Link (3 Minutes)**
  https://youtu.be/a678nAYddq0
- **Final Product Link**
  https://scam-shield-live.web.app

Scan to try the live app

Paste a suspicious message and see the verdict.



Scam Shield

Pause. Check. Then act.

Check what you were just sent before you act on it. Free, in English and Hindi.

- Gemini reads any input
- Evidence for every flag
- A scam becomes action

scam-shield-live.web.app

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

